Business Intelligence Systems

INFO 4555/5556 - Fall 2025

Data Integration



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Data Transformation and Enhancement using an "integration Platform as a Service" (iPaaS)



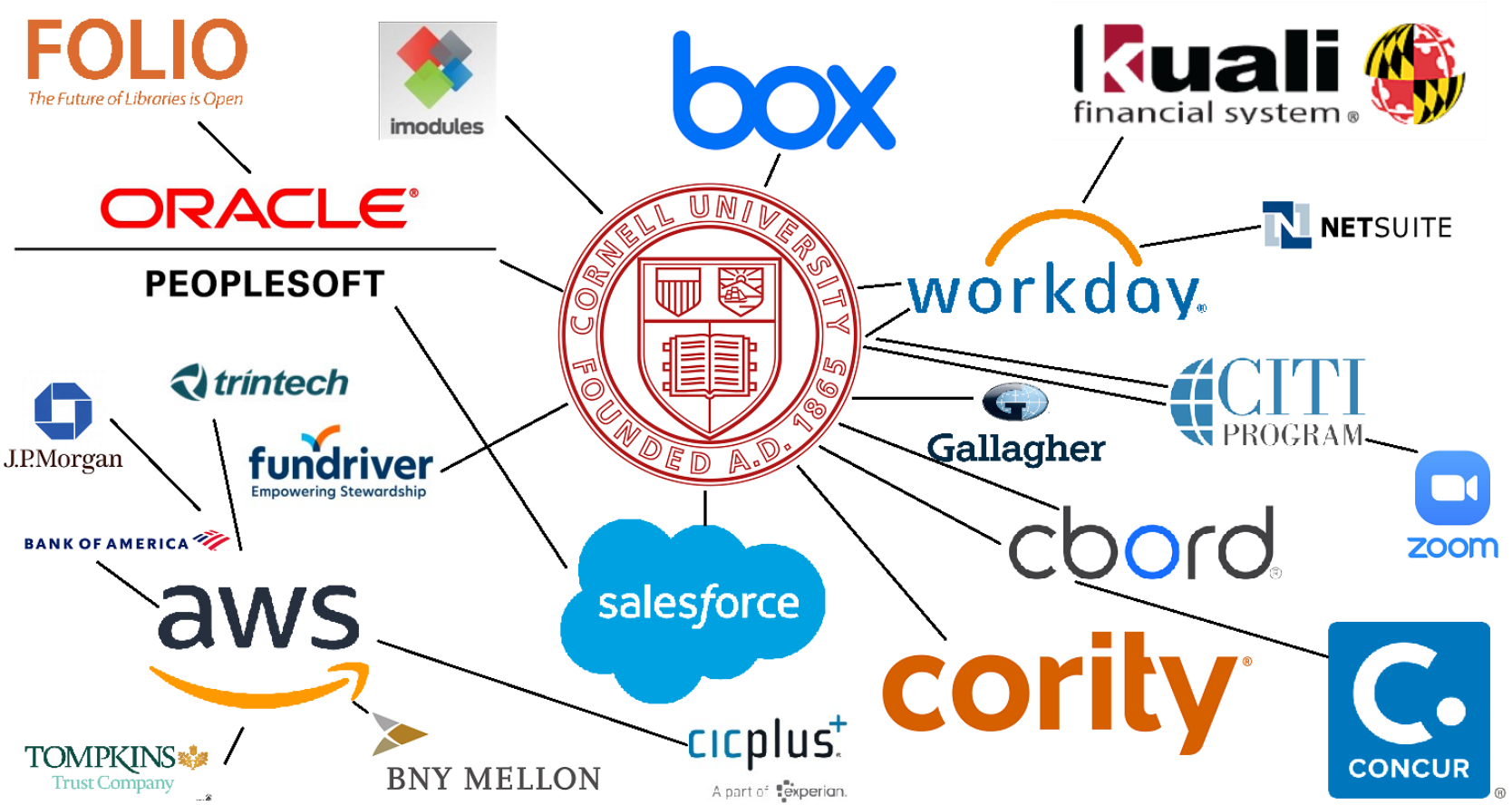
What is -aaS

- Stands for "**a**s **a** **S**ervice"
  - translates to Cloud based solutions that provide the same functionality as their on-premise equivalents
- many "**a**s **a** **S**ervice" acronyms floating around the world of "Cloud"

What is an integration

- The migration, transformation, and combination of data from disparate sources into a target system
- Ever-increasing demand for data integration as –aaS market continues to grow

What is an integration



Why use iPaaS?

- Reduce technical debt
  - Consistent design across different integrations
  - Reuse common components
  - Built-in connectors to simplify web service calls
  - Built-in version control
  - Built-in process monitoring/metrics
  - Built in process scheduling
  - Ease of use
- Conforms to latest security standards

integration Platform as a Service

iPaaS…

- Provides a cloud-based application for integrating other cloud-based or on-premise applications
- Examples include (per Gartner's 2025 Magic Quadrant Report):

  - Workato,
  - **Boomi**(),
  - Informatica,
  - SAP,
  - Microsoft,
  - Salesforce (Mulesoft),
  - Oracle,
  - IBM,
  - AWS

Source: Gartner 2025 Magic Quadrant Report

Figure 1: Magic Quadrant for Integration Platform as a Service



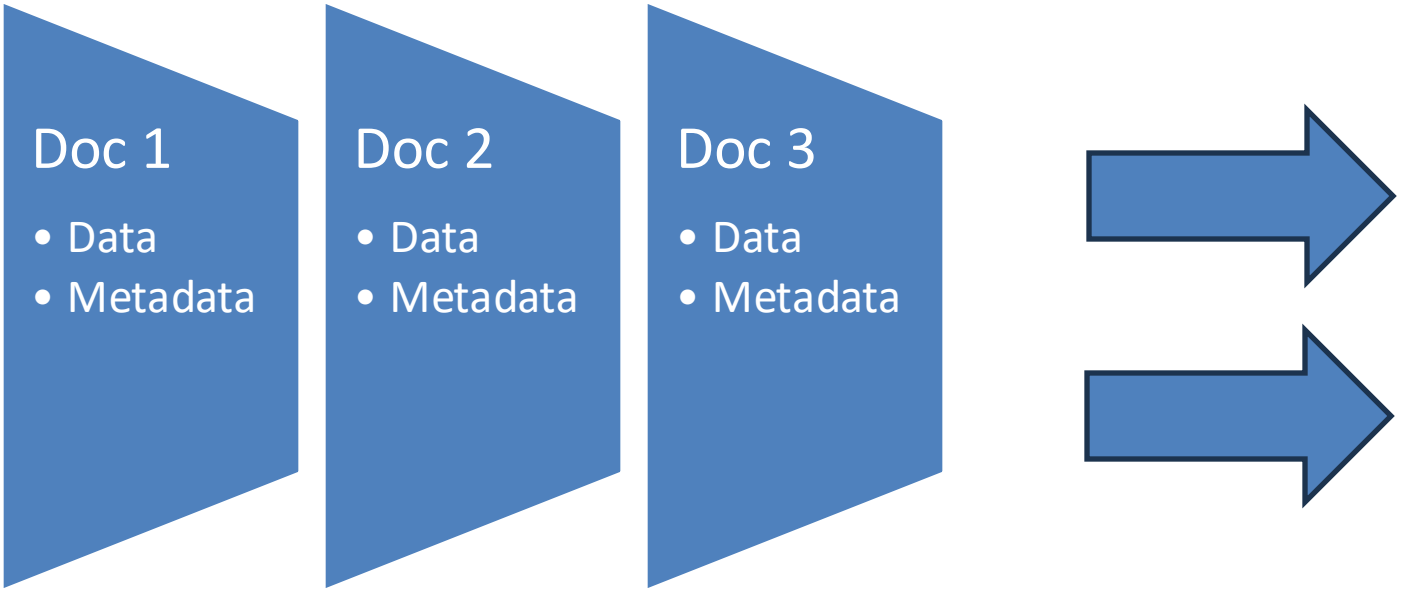
Gartner

Boomi Integrate

- A low-code point-and-click integration development platform hosted on Boomi's cloud
- Can use locally installed runtimes (small JAVA app) or Boomi's hosted runtimes to run processes
- Web-based GUI (accessible on most browsers) canvas and pre-configured shapes
- Intuitive drag-and-drop interface
- Creates a visual workflow that documents pass through using fast, multi-threaded processing

https://help.boomi.com/bundle/integration/page/int-Atoms_Molecules_and_Atom_Clouds.html

Boomi Document



Data: One unit of the information being processed

Metadata: additional information about the document, such as its (document) properties

Documents can be split, combined and processed in many ways

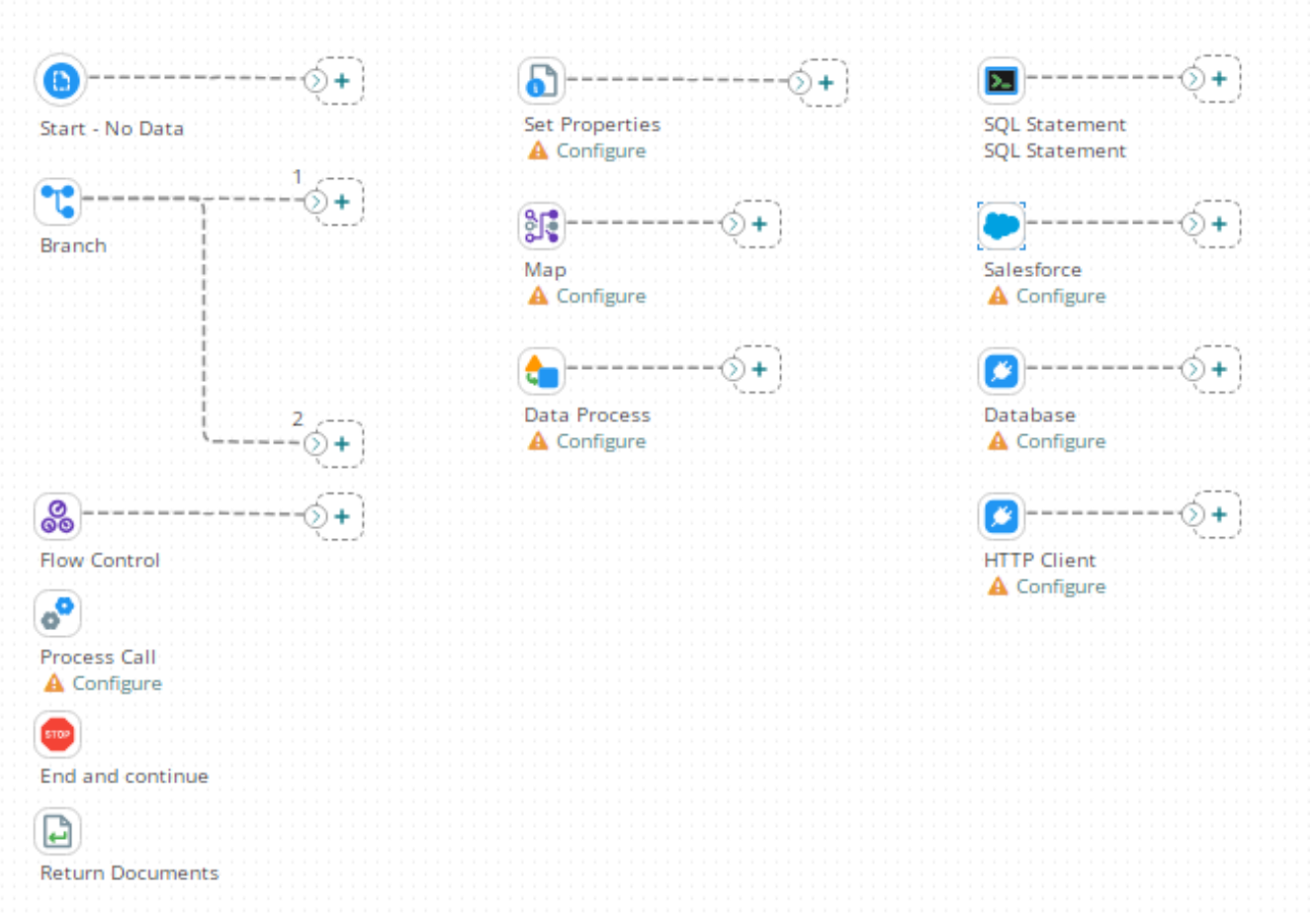
Properties

Settable within an integration to define metadata relating to full process or specific documents

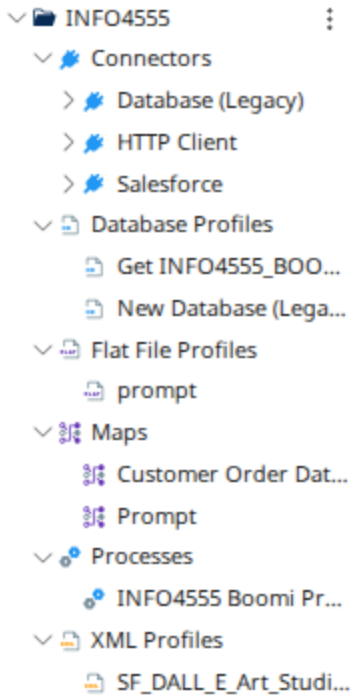
Process Properties: Properties relating to the entire process – global variables
Can be set in a process properties component, or as dynamic process properties (DPPs)

Document Properties: Properties relating to a single document – local variables
Specific document properties relating to connectors or Boomi metadata, or general dynamic document properties (DDPs)

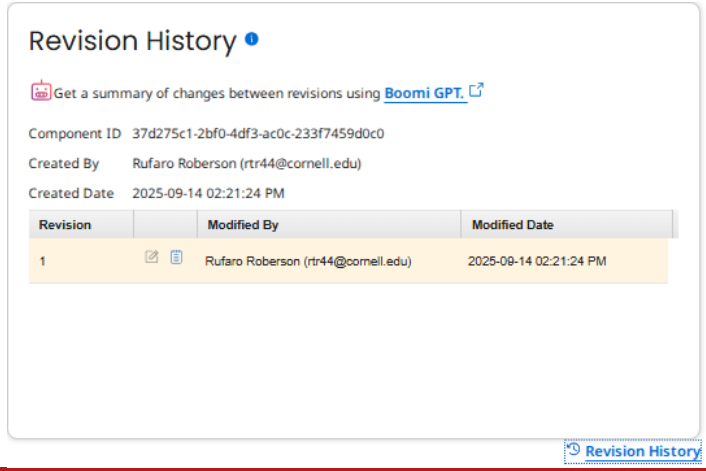
Some commonly used Boomi shapes



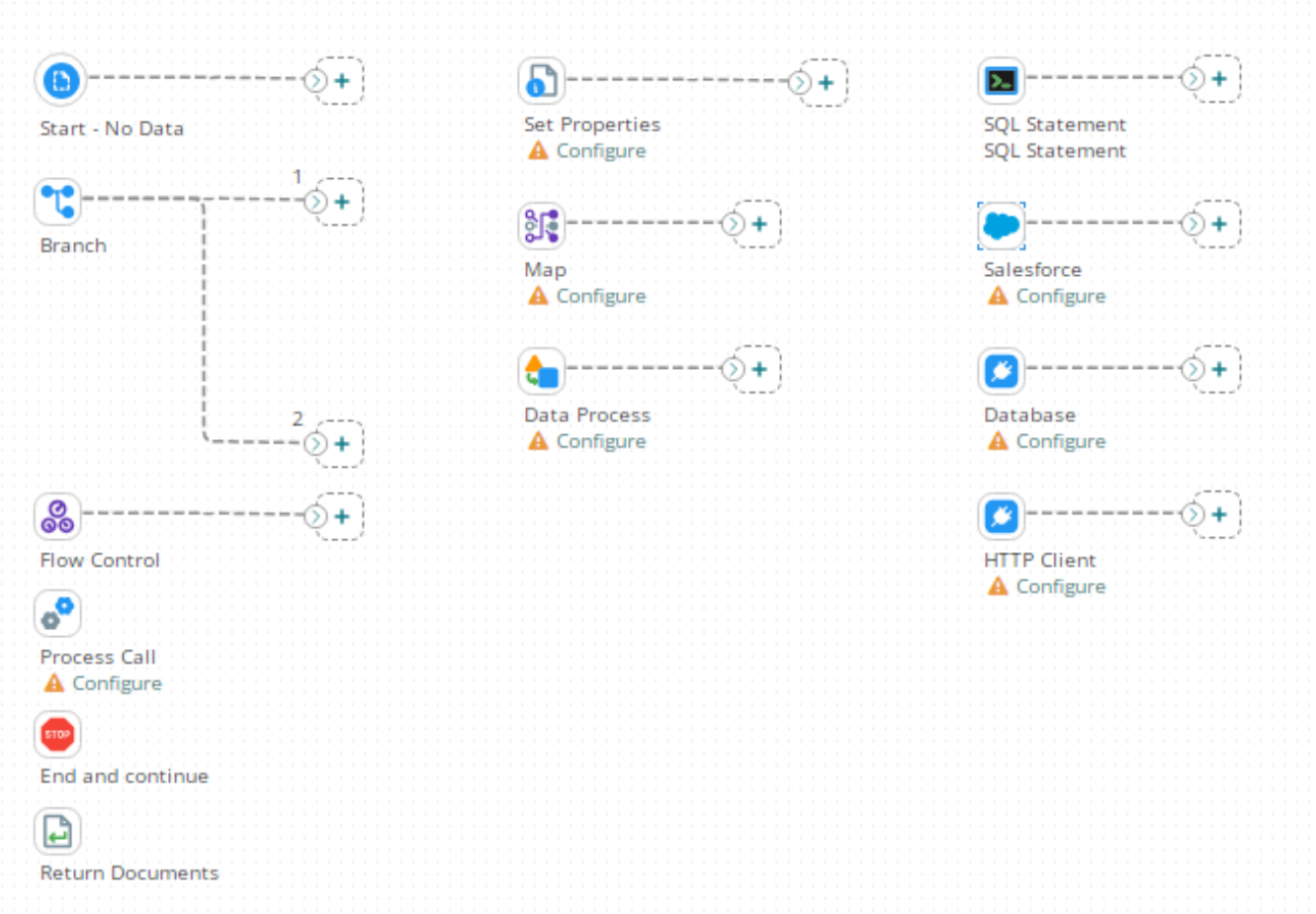
Components



- A reusable part of an integration
- Examples include connections, operations, profiles, maps, process properties, and processes
- A component may contain other components within it, e.g. a profile within a connector operation
- Can be accessed from the navigation panel on the left of the integration canvas
- You can tell whether something is a component by whether there is a 'Revision History' button in the bottom right corner

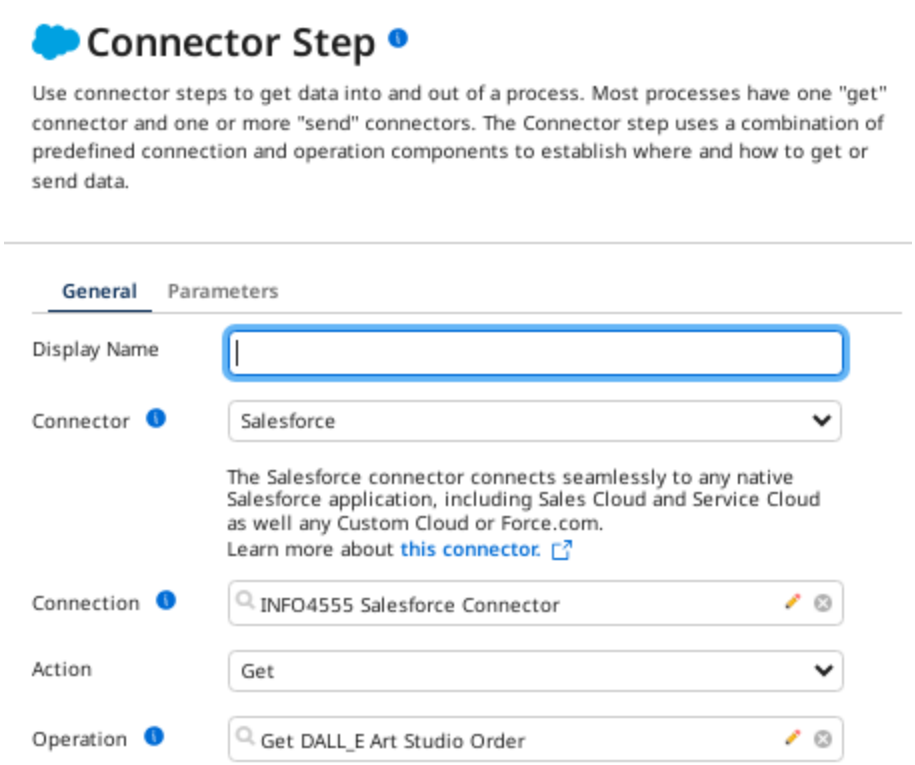


Not all shapes are components…



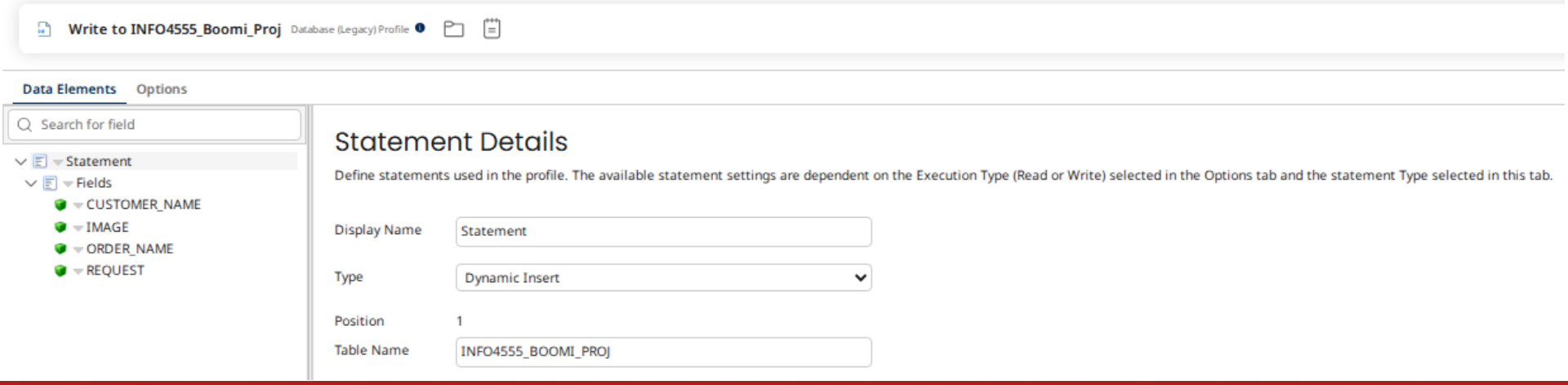
Connector Shape

- Each connector has a ‘Connection’ and ‘Operation’ component
- Connection
  - Defines basic connection information
  - Can be reused in multiple processes
- Operation
  - Specific instructions on what is being done (e.g. read or write)
  - May include profile component that can be imported from source system, or manually set, depending on connector/user needs



Profiles

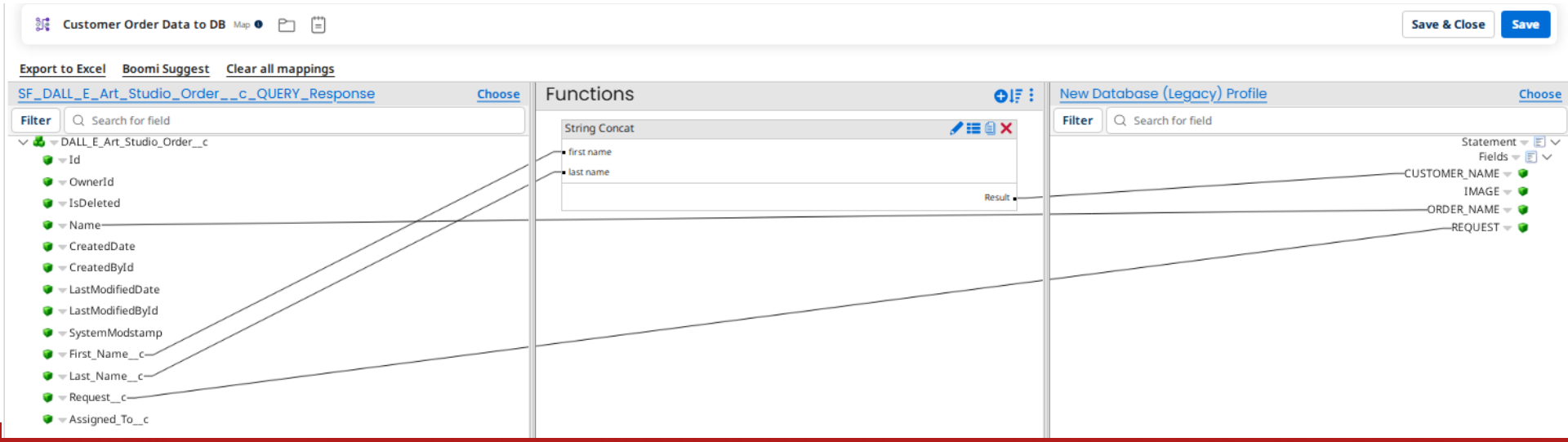
- Majority of operations include a profile that defines the exact data structure of each document; may also include additional operation information
- Profile types include database (Legacy), Flat File, JSON and XML
- Often imported or automatically created based on sample data, but can also be manually defined for simpler data structures (most commonly flat files)
- A profile is a component – the same one can be reused in multiple shapes



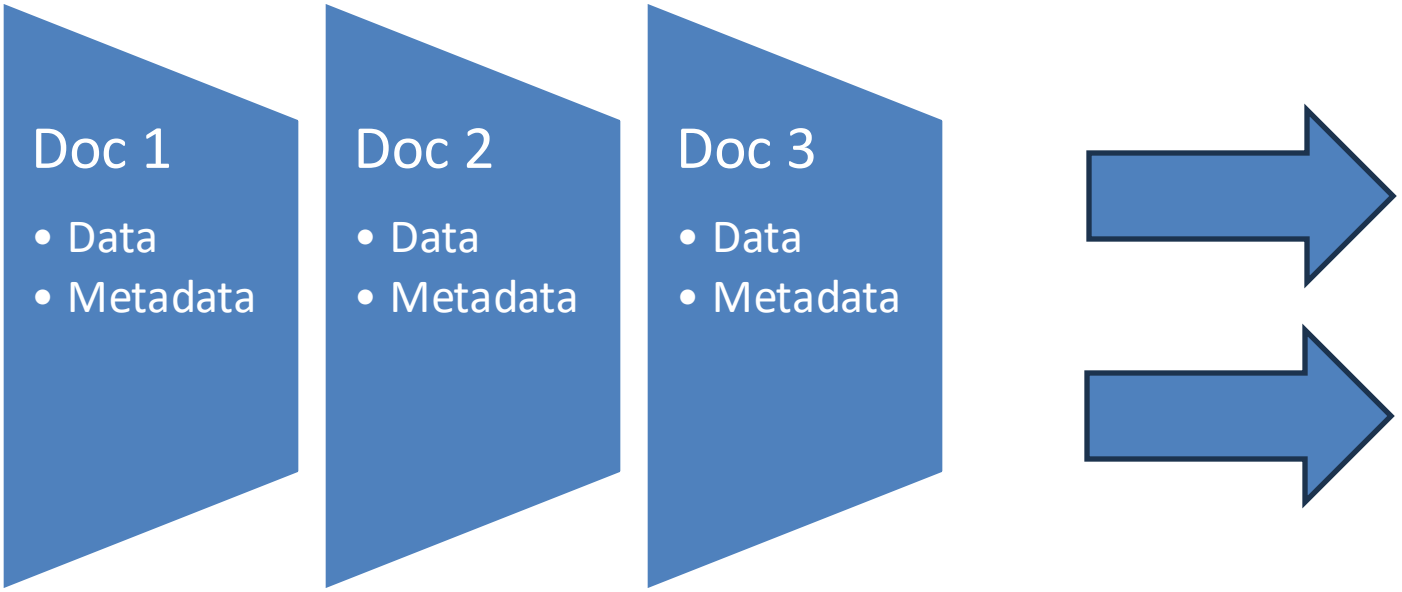
Map Shape

- Transform data from source profile (left) to target (right)
- Extra functions include string, numerical and date transformations, lookups, metadata (properties), and custom functions/scripts

Note: the profiles used in the map are the same ones used by their respective connector shape operations



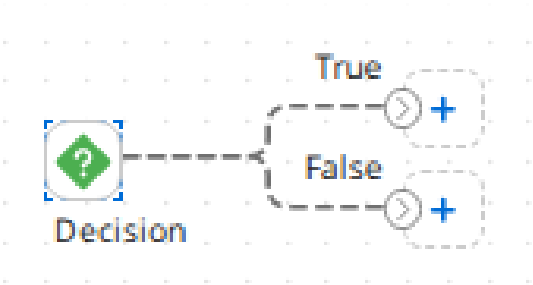
Boomi Document Flow

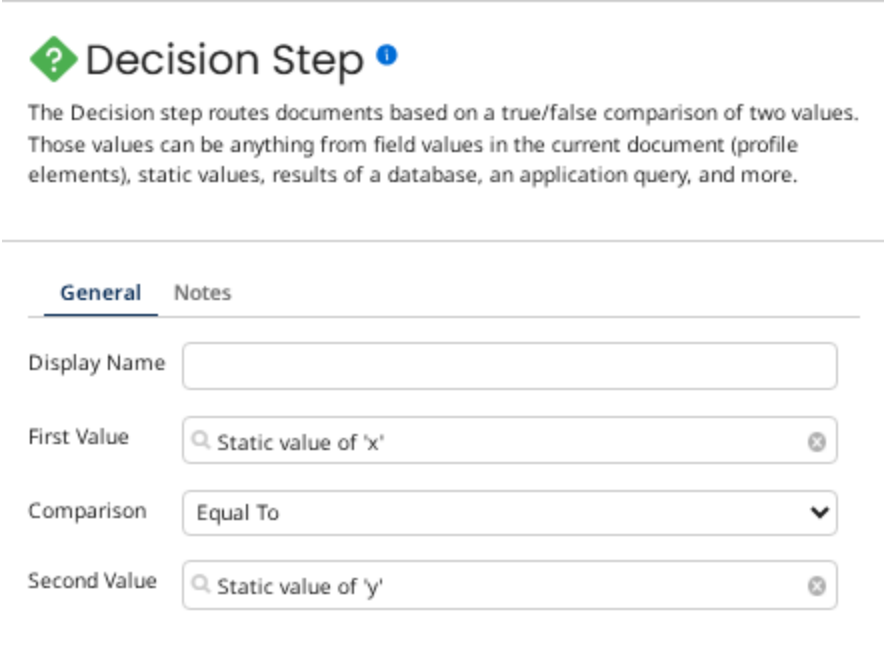


- Documents flow from shape to shape through the canvas in batches
- Processed in parallel at each step
- All documents complete processing one shape before moving on to the next
- This can be changed using a Flow Control shape

Decision Shape

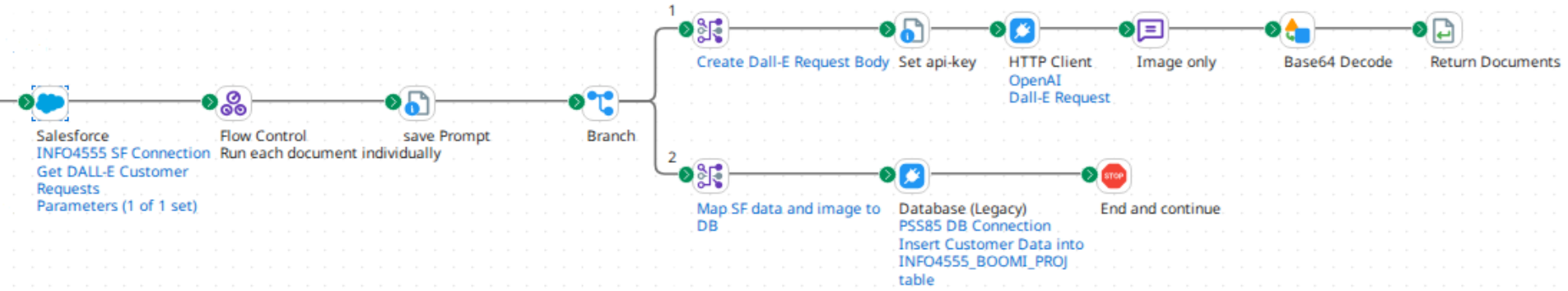
- Define a condition that evaluates to true or false, like an if-statement
- Condition can be based on document data, properties, lookups, etc.
- Documents matching the condition execute down the true path, then documents that don't match the criteria execute down the false path





Branch Shape

- Sends the same document down multiple processing pipelines
- Branches are processed in numerical order – up to 25 branches per shape
- The document state at the beginning of branch shape is preserved, i.e., any data/property changes done in one branch do not carry over to the next



Common Integration Logic

- Synchronized
  - Calculate deltas between SSoR (Source System of Record) and target system
  - Insert new records
  - Update record changes
  - Delete or deactivate records that have been deleted or deactivated in SSoR
- Flush and Fill
  - Delete all records from the target system and insert everything from SSoR

Common Integration Architectures

- Scheduled
  - Integration is set to run at specific times/time intervals

  Advantage: Can be very low technical debt, if well designed.

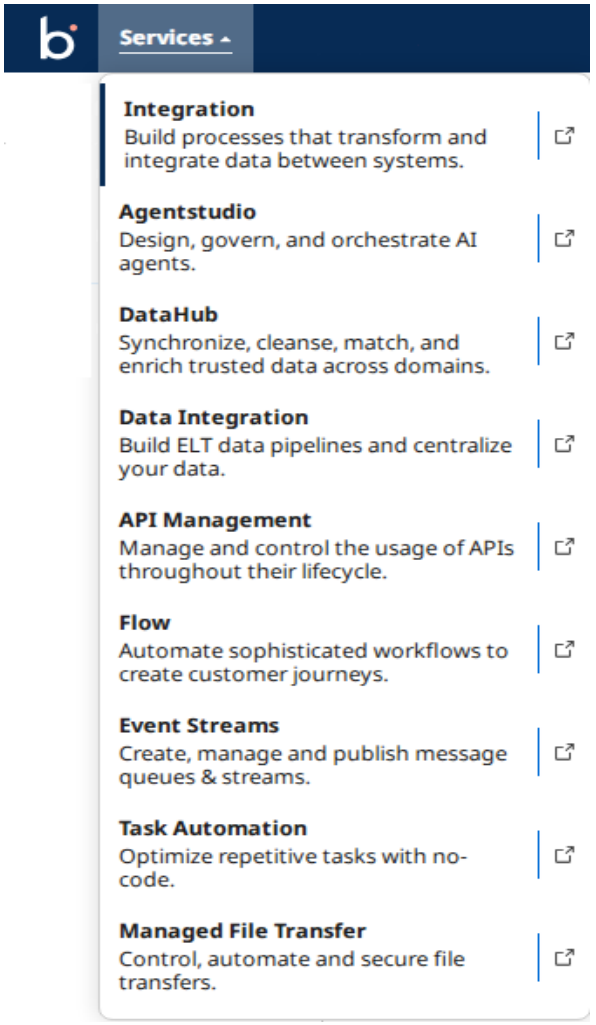
  Disadvantage: Data changes in the source system are not instantly available in the target system.

- Triggered (Realtime)
  - Integration is triggered to run instantly

  Advantage: Data is up-to-date exactly at the point when it is needed

  Disadvantage: Need to maintain system that triggers the integration and take system outages into account (e.g. add retry logic)

Other Boomi Services



Boomi API Management

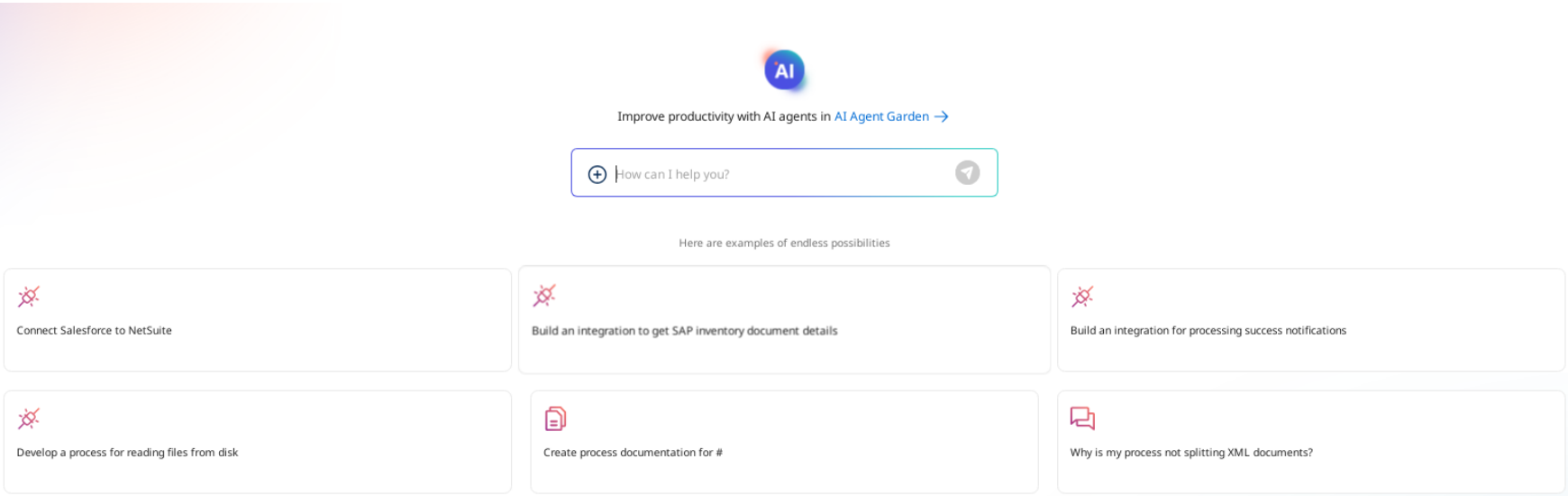
- Creates SOAP/REST/OData Web Service APIs for triggered integrations
- Allows a user without access to Boomi to run an integration using a browser, app, or code
- API calls may contain parameters that impact integration output
- APIs will be covered more in depth in Thursday's lecture

AI & the future of integrations

-Need for integration with AI systems

-Can AI build integrations?

-Boomi AI introduced Summer 2024



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Data Transformation and Enhancement using an "integration Platform as a Service" (iPaaS)

Your INFO4555/5556 Data Enhancement Assignment



Your INFO4555 Data Enhancement Lab

A common real-world requirement is to enhance data found in a source system of record (SSoR) with detail available elsewhere

In this lab exercise, you will use Boomi to build an integration that enhances data pulled from Salesforce with an image created by generative AI (OpenAI's DALL-E) using a web service call



Boomi Integration Training

- Boomi provides free training and certification in various aspects of Boomi to anyone
- Accessible at train.boomi.com
- First part of Boomi homework assignment is to complete ‘Integration Essentials’ training from the Integration Developer track
- Can optionally complete further trainings – including associate/developer certification

Your INFO4555 Data Enhancement Lab

The next few slides will provide:

- An overview of the objectives of your lab project
- A high-level flow diagram of the steps you'll need to implement
- A quick peek at the Boomi shapes you'll encounter
- Some project "rules of the road"
- Submitting proof of lab completion

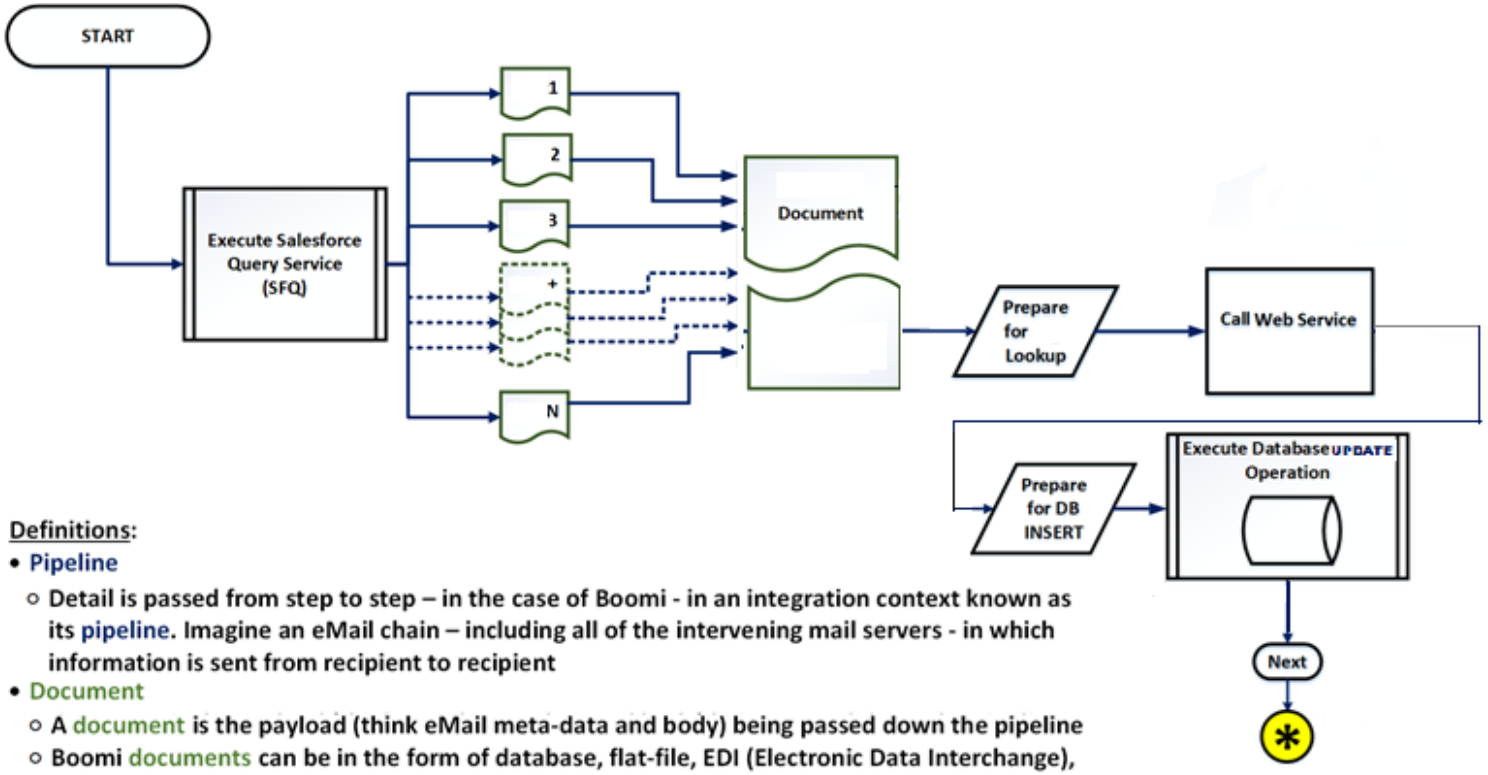


Objectives of your Lab Project

In this lab we provide an example of enhancing data via a web service (API) call. Using the Boomi integration platform, in the lab you will:

1. Extract Salesforce records pertaining to art commissions by customers
2. Feed data from Salesforce into OpenAI's DALL-E using a web service call
3. Resize and write the resulting image, and the customer data from Salesforce to a database table (INFO4555_BOOMI_PROJ)

A Flow Diagram of Your Lab's Steps

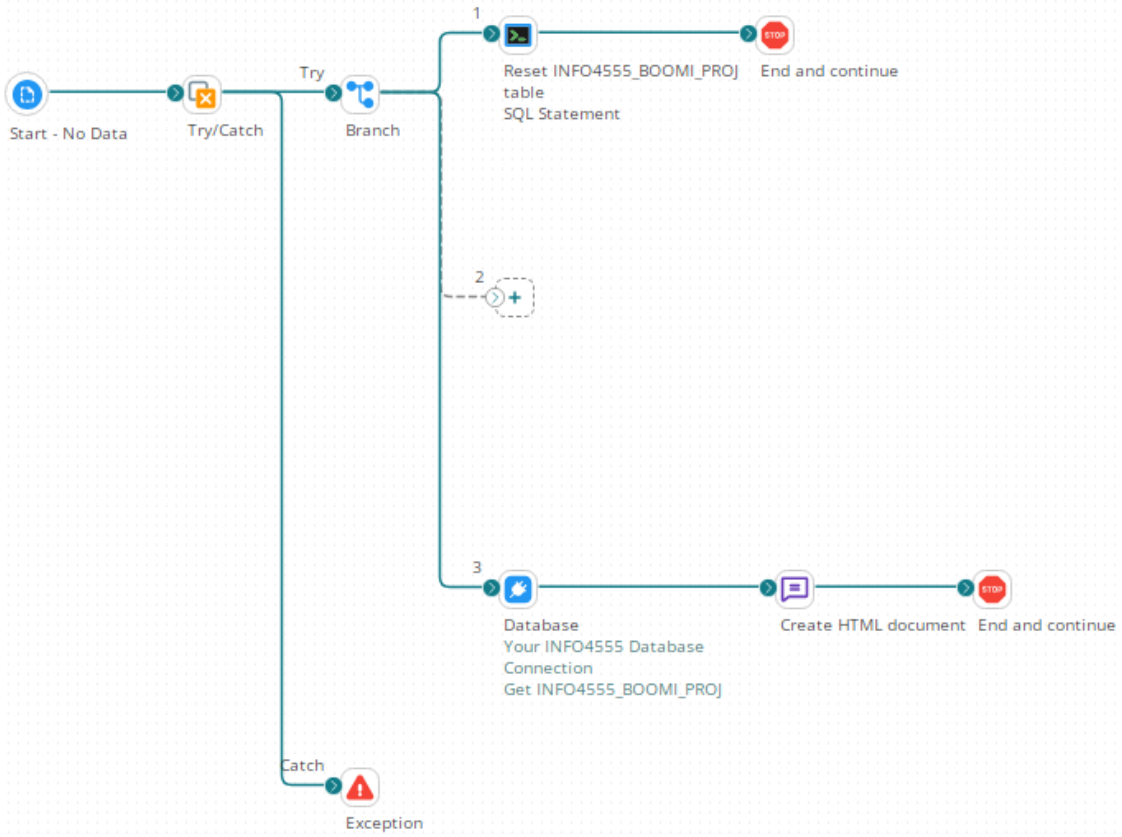


Definitions:

- Pipeline
  - Detail is passed from step to step – in the case of Boomi - in an integration context known as its pipeline. Imagine an eMail chain – including all of the intervening mail servers - in which information is sent from recipient to recipient
- Document
  - A document is the payload (think eMail meta-data and body) being passed down the pipeline
  - Boomi documents can be in the form of database, flat-file, EDI (Electronic Data Interchange), JSON or XML records
  - Most "records" take the form of elements related as parent -> child or parent -> children

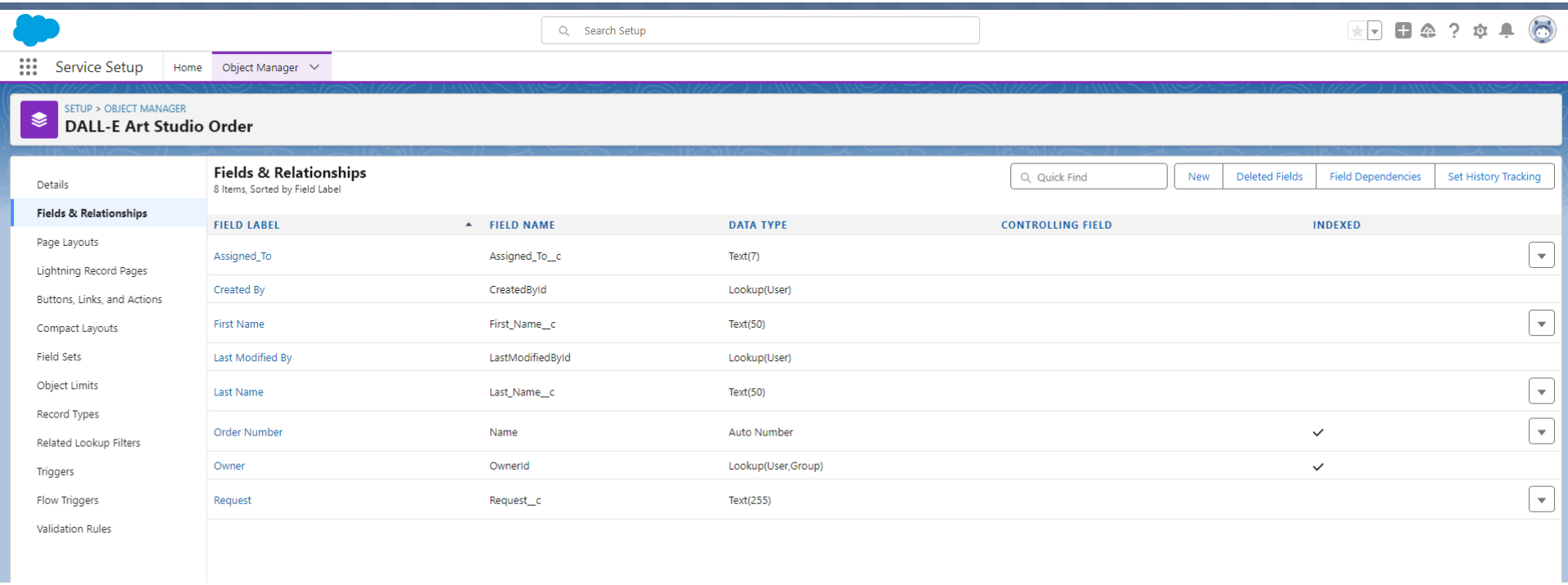
In the above example, Salesforce is returning numerous individual documents that Boomi will process in parallel unless instructed to do otherwise via a flow control device

Starting Point

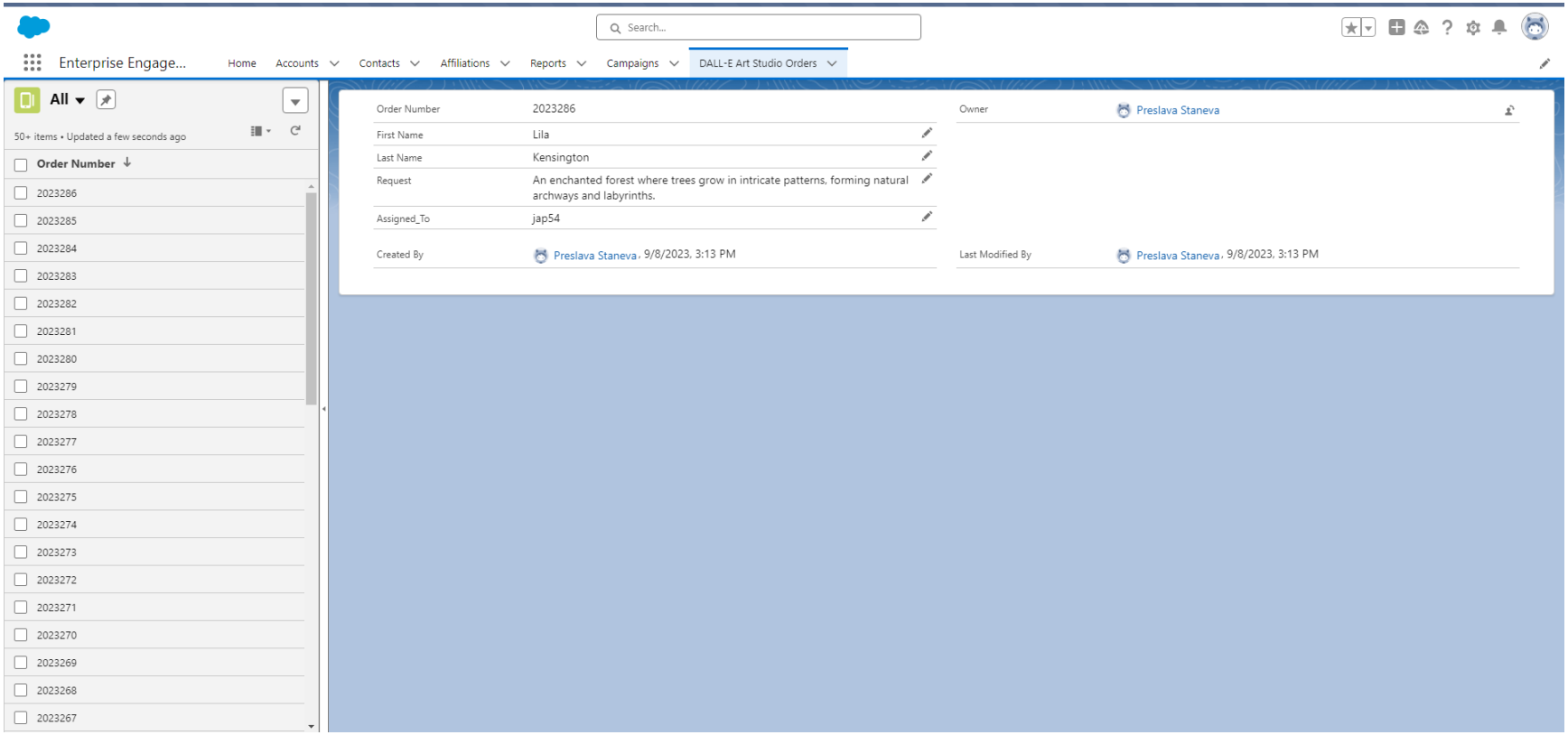


Salesforce

Custom Salesforce Object: DALL-E Art Studio Order

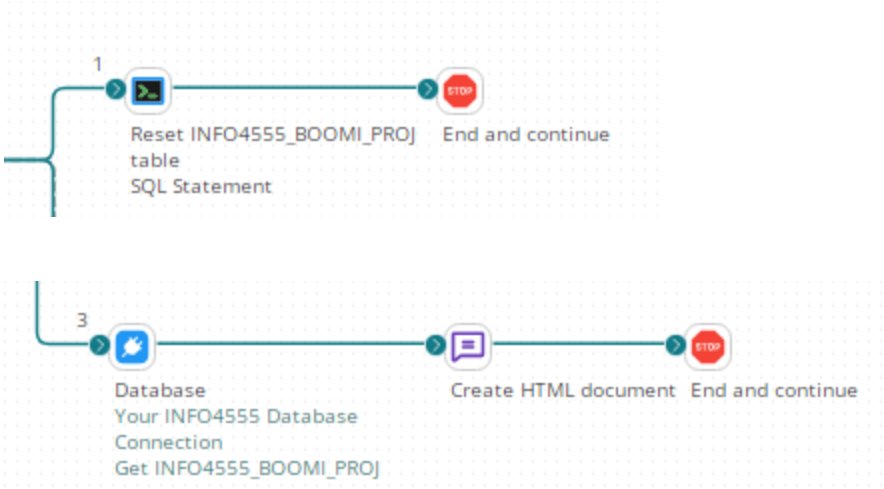


Salesforce


Search...
Enterprise Engage...
Home
Accounts
Contacts
Affiliations
Reports
Campaigns
DALL-E Art Studio Orders
All
50+ items • Updated a few seconds ago
Order Number
2023286
2023285
2023284
2023283
2023282
2023281
2023280
2023279
2023278
2023277
2023276
2023275
2023274
2023273
2023272
2023271
2023270
2023269
2023268
2023267
Order Number 2023286
First Name Lila
Last Name Kensington
Request An enchanted forest where trees grow in intricate patterns, forming natural archways and labyrinths.
Assigned_To jap54
Created By Preslava Staneva, 9/8/2023, 3:13 PM
Owner Preslava Staneva
Last Modified By Preslava Staneva, 9/8/2023, 3:13 PM

Some Lab “Rules of the Road”

- You and your classmates each have a folders in which you will develop your lab
- While only you can build or alter stuff in your folder, everyone has the ability to see and copy the contents of all other folders. **HOWEVER**, we can see who originated or has altered a component. Copying from others is a **very BIG no-no**
- After you’ve initialized your folder contents (see your lab guide for more detail), you will find a “stub” for your project. **DO NOT** alter, remove or disconnect the first or the last branches



Submitting Proof of Lab Completion

You are required to submit proof that you've completed your Boomi lab assignment. To do so, follow the instructions at the end of the assignment.

We have provided an output document in your integration that you can upload to Canvas as proof of lab completion.

Please do not alter your lab after you have submitted the worksheet to Canvas – we may run your process, and it should be in the same state as when you submitted it.

Lab Schedule and Assistance

Schedule:

- Introduction
  - Tuesday, September 23rd from 8:40 AM-9:55 AM
- Assignment Due
  - Tuesday, September 30th at 8:40 AM

Assistance:

- Lab Sections – working sessions
  - Friday, September 26th
- Ed Discussion
  - Boomi Team (Preslava, Sam, Rue) will respond to questions
- Office hours
  - Can schedule one-on-one Zoom sessions via Ed Discussion
  - May schedule Zoom office hours based on demand (see Ed Discussion for updates)